

Password Hashing Algorithms - bcrypt and Argon2

1. Why Do We Need Special Password Hashing Algorithms?

Fast hashing algorithms like:

- MD5
- SHA1
- SHA256

are not good for passwords.

Modern attackers can test billions of hashes per second using GPUs.

So password hashing algorithms are designed to be:

- slow
 - expensive
 - difficult to parallelize
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2. What is bcrypt?

bcrypt is a password hashing algorithm introduced in 1999.

It was specifically designed for securely storing passwords.

bcrypt:

- automatically uses salt

- is intentionally slow
 - supports configurable cost
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3. How bcrypt Works

Step 1

Take the password.

hello123

Step 2

Generate a random salt.

hello123 + salt

Step 3

Run multiple hashing rounds.

The number of rounds depends on the cost factor.

4. What is Cost Factor?

Cost factor controls how computationally expensive bcrypt becomes.

Example:

cost = 10

means approximately:

2^{10} iterations

Higher cost:

- more secure
 - slower computation
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5. Why bcrypt is Secure

bcrypt:

- slows down brute-force attacks
 - uses unique salts
 - adapts as hardware becomes faster
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6. Limitations of bcrypt

bcrypt is still good, but:

- not strongly memory-hard
- modern GPUs are becoming better at attacking it

This led to newer algorithms like:

- scrypt
 - Argon2
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7. What is Memory Hardness?

Memory-hard algorithms require:

- CPU power
- large amounts of RAM

This makes attacks using GPUs, ASICs, and FPGAs much more expensive.

8. What is Argon2?

Argon2 is a modern password hashing algorithm released in 2015.

It won the Password Hashing Competition (PHC).

Today, it is considered one of the best password hashing algorithms.

9. Features of Argon2

Argon2 supports:

- configurable execution time
- configurable memory usage
- configurable parallelism

This makes it highly flexible and resistant to hardware attacks.

10. Types of Argon2

Argon2d

- strongest against GPU cracking
 - vulnerable to side-channel attacks
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Argon2i

- protects against side-channel attacks
 - preferred for password hashing
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Argon2id

- hybrid version
 - combines advantages of both
 - currently the recommended choice
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11. bcrypt vs Argon2

Feature	bcrypt	Argon2
Introduced	1999	2015
Salt support	Yes	Yes
Adjustable cost	Yes	Yes
Memory hardness	Limited	Strong
GPU resistance	Good	Better
Recommended today	Acceptable	Preferred

12. Industry Recommendation

For modern systems:

- Argon2id is preferred

For older systems:

- bcrypt is still widely accepted
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13. One-Line Summary

bcrypt

A slow password hashing algorithm with built-in salt and adjustable cost.

Argon2

A modern memory-hard password hashing algorithm designed to resist GPUs and specialized cracking hardware.

Sources based on Auth0, Stytch, NordVPN, and security articles.